PRACTICAL NO:- 06

TWO SAMPLE TESTS: TWO SAMPLE RUN, KOLMOGO-ROV SMIRNOV TWO SAMLE, WILCOXON MANN WHITNEY.

(1)

$H_0: F(x) = g(x)$, The 2 sample are drawn from same distribution.

$H_1: F(x) \neq g(x)$ the two sample are not from same dis^n

Here, $n = 0$

Let $F_n(x)$ and $G_n(x)$ be the emperical with function of X and Y respectively where $F_n(x)$ is defined by

$$F_n(x) = \begin{cases} 0 & , \text{ if } x < x_{(1)} \\ i/n & , \text{ if } x_{(i)} \leq x \leq x_{(i+1)} \text{ for } i = 1, \ldots n-1 \\ 1 & , \text{ if } x \geq x_{(n)} \end{cases}$$

Similarly the emperical distributions function $G_n(Y)$ can be determined

The Kolmogorov Smirnov test statistic is given by,

$$D_{mn} = \sup_z \{|F_n(z) - G_n(z)|$$

decision rule,

If $D > D_{m,n,\alpha}$ reject H_0

x	$cum(x_i)$	$F_m(x_i)$	y	$cum(y_i)$	$G_n(y_i)$	$\lvert F_m(x) - G_n(y) \rvert$
0.6	0.6	$\frac{0.6}{20.8} = 0.0288$	2.1	2.1	$21/36 = 0.0583$	0.0295
1.2	1.8	0.0865	2.3	4.4	0.1222	0.0357
1.6	3.4	0.1634	3	7.4	0.2055	0.0421
1.7	5.1	0.2451	3.1	10.5	0.2916	0.0465
1.7	6.8	0.3269	3.2	13.7	0.3805	0.0536
2.1	8.9	0.4278	3.2	16.9	0.4694	0.0416
2.8	11.7	0.5625	3.5	20.4	0.5666	0.0041
2.9	14.6	0.7019	3.8	24.2	0.6722	0.0297
3.0	17.6	0.8461	4.6	28.8	0.8	0.0461
3.2	20.7	1	7.2	36	1	0

For $\alpha = 5\%$

If $m = n$, we combine both the observation

$\therefore n = m = 20$

$D_{(20, 0.05)} = 0.294$

$0.0536 < 0.294$

we accept H_0

$\therefore$ The two sample are drawn from same distribution.

(2)

$H_0 : F_m(x) = G(x)$, The 2 sample are drawn from same distribution

$H_1 : F_m(x) \neq G(x)$, The 2 sample are not from same distribution.

Arrange the given data in ascending order.

$m = 7, \quad n = 8$

Hours	B1	Cum(x_i)	B2	Cum(x_i)	$F_m(x_i)$	$G_n(x_i)$	$\lvert F_m(x_i) - G_n(x_i) \rvert$
70	1	1	0	0	1/7 = 0.143	0/8	0.143
80	1	2	1	1	0.286	1/8	0.161
90	1	3	0	1	0.429	1/8	0.304
100	1	4	1	2	0.571	2/8	0.32
101	1	5	0	2	0.714	2/8	0.464
115	0	5	1	3	0.714	3/8	0.339
120	0	5	2	5	0.714	5/8	0.089
125	1	6	0	5	0.857	5/8	0.232
130	1	7	1	6	1	6/8	0.25
140	0	7	1	7	1	7/8	0.125
160	0	7	1	8	1	1	0

$D_{7,8,\alpha} = 0.338$

Here, $0.338 < 0.464$
(tab)

Hence we reject H_0

$\therefore$ The 2 sample are not from same distribution

(3)

$H_0 : F(x) = G(x)$, the two sample are drawn from same distributions

$H_1 : F(x) \neq G(x)$, the two sample are not drawn from same distribution

8 14 14 14 15 18 19 19 19 19 20 21 21 22 23
Y Y Y Y X X X X X X Y Y Y X Y
1 2 3 4

24 26 26 26 26 27 28 28 29 30 30 30 31 31
Y Y X X X Y Y X X X X Y X X
5 6 7 8 9

33 34 34 35 36
X X Y Y X
10 11 12

Runs $= R = 12$ $\qquad n_1 = 18, \; n_2 = 16$

$R^{(0.05)}_{(18,16)} = 11$

$$E(R) = \frac{2n_1n_2}{n_1+n_2} + 1 = \frac{2(18)(16)}{18+16} + 1 = 17.9411$$

$$V(R) = \frac{2n_1n_2(2n_1n_2 - n_1 + n_2)}{(n_1+n_2)^2(n_1+n_2-1)} = \frac{2\times18\times16(2\times18\times16-18+16)}{(18+16)^2(18+16-1)}$$

$V(R) = 8.667$

$$Z = \frac{R - E(R)}{\sqrt{V(R)}} \sim N(0,1), \text{ Reject } H_0 \text{ if } |Z| > Z_{\alpha/2}$$

$$= \frac{12 - 17.9411}{\sqrt{8.667}} = \frac{-5.9411}{2.9439} = -2.0181$$

$|Z| = 2.0181$

$\therefore 2.0181 > 1.96$

Hence we reject H_0

$\therefore$ The two sample are not drawn from same distribution.

(4) $H_0 : F(x) = G(x)$, the two sample are drawn from same distribution

$H_1 : F(x) \neq G(x)$, the two sample are not drawn from same distribution.

10	15	17	21	22	22	22	24	25	27	29	30	30	36
Y	Y	Y	Y	Y	Y	X	X	X	X	X	X	Y	X

38	38	41	45	49	49	50	55	55	58	60	60	61	69	70	72
Y	X	X	Y	Y	X	Y	Y	X	X	X	Y	Y	X	X	X
3	4		5		6	7		8			9		10		

$R = 10$, $n_1 = 15$, $n_2 = 15$

$$E(R) = \frac{2 \times 15 \times 15}{15+15} + 1 = 16$$

$$V(R) = \frac{2 \times 15 \times 15 (2 \times 15 \times 15 - 15 + 15)}{(15+15)^2 (15+15-1)} = 7.7586$$

$$Z = \frac{10-16}{\sqrt{7.7586}} = -2.1540$$

$|Z| = 2.154$

$\therefore 2.154 > 1.96$

$\therefore$ we reject H_0

Hence the 2 sample are not from same distribution

(5) H_0: There is a significante amongst the scores in two categories.

H_1: There is no significance among the scores in two categories.

By using Mann whitney - U test

$n_1 = n_2 = 5$

we define $\phi(x_i, y_i)$ {We use this test only when sample size is less than 8}

$$\phi(x_i, y_i) = \begin{cases} 1, & \text{if } x_i < y_i \\ 0, & \text{if } x_i > y_i \end{cases}$$

$$u_k = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \phi(x_i, y_i) \quad , k = 1, 2 \ldots$$

$$u = \min(u_1, u_2)$$

Thus u_1 measures the number of times x_i precedes y_j amongst all possible pairs of (x_i, y_j) and u_2 measures the number of times y_j precedes x_i amongst all possible pairs of x_i, y_j

Combine both and write in ascending order,

y_1	y_2	y_3		y_4	y_5				
10	28	33	48	52	60	89	90	110	125
B	B	B	A	B	B	A	A	A	A

Here $u_1 = 2$

For $u_2 =$ total combination $- u_1$

$= 25 - 2 = 23$ $[3+5+5+5+5 = 23]$.

$\min(u_1, u_2) = \min(2, 23) \Rightarrow 2$

Decision rule,

$u > u_{n,\alpha}$ we accept H_0

$u \leq u_{n,\alpha}$ we reject H_0

$$u_{(5,5)}^{(0.05)} = 2$$

$\therefore 2 = 2$

$\therefore$ we reject H_0

Hence, there is no significance among the scores in two categories.

(6)

H_0 : The Scores of English group differ significantly with that of mother tongue group.

H_1 : The Scores of English group donot differ significantly with that of mother tongue group

		Scores	Rank
E	1	10	1.5
M	2	10	1.5
E	3	11	3.5
M	4	11	3.5
M	5	12	4.5
M	6	12	4.5
E	7	13	8
M	8	13	8
M	9	13	8
E	10	15	10.5
M	11	15	10.5
E	12	16	12.5
M	13	16	12.5
E	14	17	15
E	15	17	15
M	16	17	15
E	17	18	17

T_1 = sum of ranks of english $\Rightarrow T_1 = 83$
T_2 = sum of ranks given to mother tounge $\Rightarrow T_2 = 70$

$$u_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - T_1$$

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$$= 8 \times 9 + \frac{8(8+1)}{2} - 83$$

$$\boxed{U_1 = 25}$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - T_2$$

$$= 8 \times 9 + \frac{9(9+1)}{2} - 70$$

$$\boxed{U_2 = 47}$$

$$U = \min(U_1, U_2) \Rightarrow \min(25, 47) \Rightarrow 25$$

$$\boxed{U = 25}$$

$$U^{(0.05)}_{(8,9)} = 15$$

$$25 > 15$$

$\therefore$ we accept H_0

Hence, the scores of English group differ significantly with that of Mother tounge group.

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